What should an analyst look for in DNS, HTTP, HTTPS/TLS, ARP, ICMP and SMB traffic?

Table: protocol, normal behaviour, suspicious behaviour.

Network traffic analysis is the process of capturing, examining, and interpreting network packets to understand communication patterns, troubleshoot network problems, and detect security threats.

A security analyst examines source and destination addresses, ports, protocols, packet contents where available, timing, and communication frequency.

1. Protocol Analysis Table

Protocol	Normal Behaviour	Suspicious Behaviour
DNS	Resolves domain names into IP addresses using DNS queries and responses	DNS tunneling, unusually long subdomains, excessive DNS queries, random-looking domains, high NXDOMAIN rates
HTTP	Transfers web requests and responses using methods such as GET and POST	SQL injection attempts, malicious uploads, suspicious URLs, unusual POST requests, webshell access
HTTPS/TLS	Establishes encrypted communication and authenticates servers through certificates	Invalid certificates, unexpected destinations, weak TLS configurations, unusual TLS fingerprints, abnormal data transfers
ARP	Resolves IPv4 addresses to MAC addresses within a local network	ARP spoofing, conflicting IP-to-MAC mappings, excessive unsolicited ARP replies
ICMP	Provides network diagnostics, error reporting, and connectivity testing	ICMP flooding, ping sweeps, unusually large payloads, ICMP tunneling
SMB	Enables file sharing, printer sharing, and access to shared network resources	Unauthorized share access, repeated login failures, lateral movement, unusual file transfers

2. DNS Traffic Analysis

DNS (Domain Name System) translates domain names into IP addresses so devices can locate network services.

DNS commonly uses UDP port 53, although TCP port 53 is also supported.

Normal DNS communication involves a client sending a query to a DNS resolver and receiving a response containing the requested record or an error.

Important DNS records include:



Record	Purpose
A	Maps a hostname to an IPv4 address
AAAA	Maps a hostname to an IPv6 address
MX	Identifies mail servers
NS	Identifies authoritative name servers
CNAME	Creates an alias for another domain name
TXT	Stores text-based information
PTR	Supports reverse DNS lookups

Suspicious DNS Indicators

- Excessive queries from a single endpoint.
- Randomly generated or unusually long domain names.
- Frequent NXDOMAIN responses.
- High volumes of TXT record queries.
- DNS communication with unauthorized resolvers.
- Repeated queries at regular intervals.
- Encoded data within subdomains.

These indicators may suggest malware communication, domain generation algorithms (DGAs), or DNS tunneling.

3. HTTP Traffic Analysis

HTTP (Hypertext Transfer Protocol) is an application-layer protocol used to exchange web resources between clients and servers.

HTTP commonly uses TCP port 80.



Important HTTP methods:

Method	Purpose
GET	Requests a resource
POST	Submits data to a server
PUT	Creates or replaces a resource
DELETE	Requests deletion of a resource
HEAD	Retrieves response headers without the response body
OPTIONS	Identifies supported communication options

Important HTTP status codes:

Status Code	Meaning
200	OK
301	Moved Permanently
302	Found / Temporary Redirect
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Internal Server Error
503	Service Unavailable

Suspicious HTTP Indicators

- Repeated failed login requests.
- SQL injection patterns in URL parameters.
- Cross-site scripting payloads.
- Unexpected file uploads.
- Downloads of suspicious executable files.
- Unusual User-Agent strings.
- Requests to sensitive paths such as /admin, /backup, or /config.
- Unexpected increases in HTTP 4xx or 5xx responses.

4. HTTPS/TLS Traffic Analysis

HTTPS (Hypertext Transfer Protocol Secure) uses TLS to protect HTTP communication through encryption, integrity protection, and server authentication.

HTTPS commonly uses TCP port 443. HTTP/3 uses QUIC over UDP, commonly on port 443.

TLS communication generally involves:



1. ClientHello - Client proposes TLS parameters.
2. ServerHello - Server selects negotiated parameters.
3. Authentication and key establishment - Parties establish secure keys and authenticate the server.
4. Encrypted communication - Application data is exchanged securely.

In TLS 1.3, much of the handshake after ServerHello is encrypted.

Important Analysis Fields

- Source and destination IP addresses.
- TLS protocol version.
- Cipher suites.
- Certificate validity and issuer, when observable.
- Server Name Indication (SNI), when visible.
- TLS fingerprint information.

- Connection frequency, duration, and data volume.

Encrypted application contents generally cannot be examined without authorized decryption.

5. ARP Traffic Analysis

ARP (Address Resolution Protocol) maps IPv4 addresses to MAC addresses on a local network.

It operates between network-layer addressing and link-layer communication.

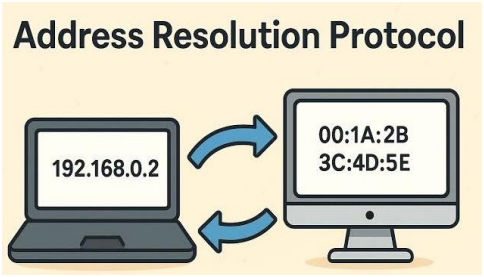
Example:

ARP Request:

Who has 192.168.1.1?

ARP Reply:

192.168.1.1 is at AA:BB:CC:DD:EE:FF



Suspicious ARP Indicators

- Multiple MAC addresses claiming the same IP address.
- Unexpected changes in the default gateway's MAC address.
- Excessive gratuitous or unsolicited ARP messages.
- Conflicting ARP responses.

These activities may indicate ARP spoofing or a man-in-the-middle attack, although legitimate network changes can produce similar symptoms.

6. ICMP Traffic Analysis

ICMP (Internet Control Message Protocol) is used for network diagnostics and error reporting. It is commonly associated with the ping and traceroute utilities.

Important ICMPv4 messages:

Type	Description
0	Echo Reply
3	Destination Unreachable
5	Redirect
8	Echo Request
11	Time Exceeded



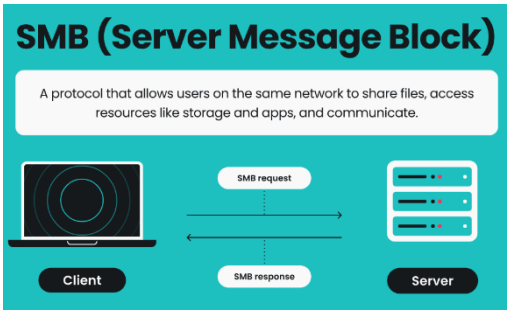
Suspicious ICMP Indicators

- High volumes of ICMP Echo Requests.
- Ping sweeps across multiple hosts.
- Unusually large ICMP payloads.
- Repeated ICMP communication with external hosts.
- Encoded data within ICMP payloads.

These patterns may indicate host discovery, denial-of-service activity, or ICMP tunneling.

7. SMB Traffic Analysis

SMB (Server Message Block) is an application-layer protocol used for network file sharing, printer sharing, and remote access to shared resources.



Modern SMB commonly uses TCP port 445.

Important SMB concepts:

- SMB Sessions: Authenticated connections between clients and servers.
- Shares: Network-accessible resources.
- Administrative Shares: Special shares such as C$, ADMIN$, and IPC$.
- SMB Signing: Helps protect message integrity and reduce certain relay and tampering risks.
- SMB Encryption: Protects SMB data confidentiality when enabled.

Suspicious SMB Indicators

- Unexpected access to administrative shares.
- Repeated authentication failures.
- Connections between unusual workstation pairs.
- Large-scale file modification or encryption.
- Unexpected copying of executable files.
- Attempts to access sensitive shared folders.

Such behavior may indicate lateral movement, credential misuse, or ransomware activity.